

Contents

Routing Attention Through Memory: A Recurrent Vision Agent Solves Spatial and Value-Guided Change Detection by Attending to Its Own Recurrent State	1
Abstract	1
1. Introduction	1
2. Methods	4
3. Results	7
4. Discussion	16
5. Conclusion	19

Routing Attention Through Memory: A Recurrent Vision Agent Solves Spatial and Value-Guided Change Detection by Attending to Its Own Recurrent State

A conv-free recurrent agent learns Posner-style cued change detection, and its decisions are causally carried by cross-attention onto memory tokens rather than the live image.

Abstract

Attention in biological vision is guided not only by the incoming image but by internal expectations about where and how much a stimulus matters. We ask whether an artificial agent, trained end-to-end by reinforcement learning on a demanding perceptual decision, discovers an analogous internal organization. We study a conv-free recurrent vision agent that perceives each frame as a set of patch tokens, updates them through a single cross-attention block whose keys and values are the patch tokens concatenated with two recurrent memory states carried as tokens, and reads out actions and a distributional value from the top memory through small one-dimensional convolutions. Trained purely by reinforcement learning on a Posner-style cued change-detection task, the agent reproduces the hallmark signatures of covert spatial attention: a clean psychometric function rising to a true ceiling, faster and more accurate detection at cued than uncued locations with a validity benefit that peaks at intermediate change magnitudes and vanishes at ceiling, and a value-graded modulation by cue colour. Dissecting the trained network, we find that its perceptual cross-attention is overwhelmingly directed at the recurrent memory rather than the live image: roughly ninety percent of each patch query’s attention falls on the two memory states, and these attention maps track the cued and changed quadrants. Cross-validated decoders show that cue identity, reliability and elapsed time are linearly available from the latents, and that the location of the change is held in the spatial layout of memory rather than its pooled average. Causal biasing of the attention logits confirms that the decision is carried by attention onto memory, not onto the image. The critic is near-perfectly calibrated and orders its value estimates by expected reward, with distributional uncertainty that collapses as evidence accrues. Together these results identify recurrent memory as the substrate through which a learned agent routes spatial and value-directed attention.

1. Introduction

Selective visual attention is the brain’s solution to a structural bottleneck: the visual field delivers far more information than can be deeply processed or acted upon, so the system must continuously

decide which locations and which features deserve scrutiny. A central insight from decades of psychophysics is that this selection is steered by internal signals — a prior expectation about where a relevant event will occur, and a valuation of how much that event is worth — and not solely by the salience of the current image. The Posner cueing paradigm crystallized this idea: an informative spatial cue presented before a target speeds and sharpens responses at the cued location (a validity benefit) at the cost of responses elsewhere, demonstrating that attention can be allocated covertly, in advance of the stimulus, on the basis of an internal expectation. Layered on top of spatial expectation is value: when locations or features differ in the reward they predict, attention and the decisions it supports are biased toward the more valuable option, linking perception to the machinery of reinforcement.

These phenomena are typically studied in biological observers or modeled with hand-built attentional mechanisms. It is far less clear whether, and how, an artificial agent that is given no attentional prior and no explicit notion of a cue will nonetheless discover spatial and value-directed attention purely as a consequence of having to act well under reward. If it does, the trained network becomes a concrete, fully observable object in which we can ask a mechanistic question that is difficult to pose in the brain: where, physically, does the attentional prior live, and through what computation does it reach the decision?

Here we train a recurrent vision agent end-to-end by reinforcement learning on a Posner-style cued change-detection task (Fig. 1) and then dissect the trained model. The agent watches a short movie in which a coloured spatial cue — whose colour signals reward magnitude and whose appearance signals reliability — precedes the onset of four oriented patches in the four quadrants; at an unpredictable moment one patch changes orientation, and the agent must press to report the change. The task therefore jointly probes spatial expectation (valid versus invalid changes relative to the cue), perceptual sensitivity (graded change magnitude), and value (reward-signalling cue colour) within a single closed-loop control problem.

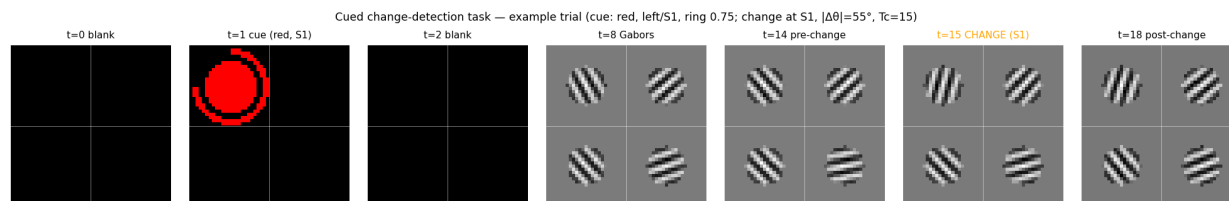


Figure 1: Task schematic. A 29-frame Posner-style cued change-detection trial: an initial blank, a coloured spatial cue (a ring whose colour signals reward magnitude and whose proportion signals reliability), a blank delay, then four Gabor patches in the four quadrants (S1 top-left, S2 bottom-left, S3 top-right, S4 bottom-right). At a random change time one patch steps in orientation by $|\Delta\theta|$; the agent presses to report it. Valid trials change at the cued quadrant, invalid trials elsewhere.

Our central finding is that the agent solves this task by routing its attention through its own recurrent memory rather than through the live image. The network’s only perceptual attention operation is a cross-attention block in which the current patch tokens query a key/value set built from the patches together with two recurrent memory states carried as tokens. We show that the trained agent directs the overwhelming majority of this attention onto the memory tokens, that the memory carries a spatially-resolved representation of the cued and changed locations, and — by causally perturbing the attention logits — that the decision is moved by attention onto memory and is essentially unaffected by attention onto the image. We then characterize the value system that drives behaviour: a near-perfectly calibrated distributional critic whose value estimates are ordered by

expected reward and whose uncertainty collapses as perceptual evidence accumulates.

This paper makes five contributions. (1) Behaviour: the agent reproduces the psychometric, chronometric, spatial-validity and value-grading signatures of covert and value-directed attention. (2) A memory-dominated perceptual attention: its cross-attention places roughly ninety percent of its weight on recurrent memory, with sharp, spatially-organized maps that follow the cue and lock onto the change. (3) What the latents encode: cue identity, reliability and elapsed time are linearly decodable, and — critically — the location of the change is represented in the spatial layout of memory rather than its pooled summary. (4) A causal locus of decision: biasing attention onto memory moves the decision while biasing attention onto the image does not. (5) A calibrated, value-directed critic: the distributional value function is unbiased relative to realized return, ordered by reward, and resolves its uncertainty as evidence grows. Together these establish recurrent memory as the medium through which a learned agent composes spatial expectation, evidence and value into action.

1.1 Background and related work

Covert spatial attention and the Posner cue. The behavioural hallmark of endogenous spatial attention is the validity effect: a predictive pre-cue improves the speed and accuracy of responses to targets appearing at the cued location relative to uncued locations. This benefit is classically interpreted as a prior expectation that pre-allocates processing resources to a location before any target is present, and it is strongest when the target is hard to detect and disappears when the target is so salient that performance saturates — a signature that any model claiming to capture covert attention should reproduce. The paradigm has become a standard assay precisely because it dissociates stimulus-driven from expectation-driven selection.

Value-directed attention. A second body of work shows that attention is not value-neutral: stimuli and locations associated with larger reward are prioritized, and this prioritization is shaped by reinforcement history rather than by physical salience alone. This connects attention to reinforcement learning, where the value of acting must be estimated from experience, and motivates studying attention inside an agent that is trained by reward and that maintains an explicit value estimate.

Attention mechanisms in artificial vision and recurrent computation. Modern artificial vision relies heavily on learned attention, in which a set of query representations is compared against keys to produce a content-addressed weighting over values. In most vision systems this attention operates over representations of the current input. A complementary tradition in recurrent and memory-augmented models treats internal state as something that can itself be addressed and read, allowing a network to attend to what it has stored rather than only to what it currently sees. The agent studied here sits at this intersection: a single cross-attention block in which patch tokens may attend either to the live image or to two recurrent memory states represented as tokens, with the memory carried across frames and fed back as the keys and values. This design makes the allocation of attention between image and memory an empirical, measurable quantity rather than an architectural assumption — and it is exactly this allocation that our analyses exploit.

Mechanistic interpretability of trained agents. Finally, our approach is interpretive: rather than proposing an attentional mechanism by hand, we train an agent under reward and then ask what mechanism it discovered, using attention-weight analysis, cross-validated linear decoding of task variables from internal states, causal perturbation of internal computations, and calibration of the learned value function. The contribution is to bring these tools to bear jointly on a single agent performing a task with well-understood psychophysics, so that the discovered mechanism can be

evaluated against the established phenomenology of spatial and value-directed attention.

2. Methods

2.1 The change-detection environment

The agent was trained and probed on a Posner-style cued change-detection task. Each trial is a sequence of 29 RGB frames at 50×50 pixels, with pixel intensities in $[-1, 1]$. Frames unfold in four phases. The trial opens with one or more blank frames; a single coloured spatial cue then appears as a ring in one of two locations; after a further blank the stimulus display comes on, showing four oriented Gabor patches, one in each quadrant — S1 (top-left), S2 (bottom-left), S3 (top-right), S4 (bottom-right) — each occupying a 25×25 region of the image and replicated across the three colour channels. The four base orientations are drawn uniformly and per-frame orientation noise is added, so the patches flicker around their means.

The cue carries two task-relevant signals. Its colour encodes the reward that a correct report will earn — red = 5, green = 3, blue = 1 — and the angular completeness of the ring (its “proportion”, taking values 0.25, 0.5, 0.75, 1.0) is a displayed reliability signal. The cue also points to a side: a left cue is associated with S1 and a right cue with S4. With probability equal to the ring proportion the impending change occurs at the cued quadrant; otherwise it occurs at one of the other three quadrants chosen at random.

At a change time drawn uniformly from frames 11–25, one Gabor’s orientation steps by a fixed magnitude $|\Delta\theta|$ (the change magnitude per trial, drawn in $[-\theta, \theta]$); the agent’s task is to detect this orientation change and report it by emitting a press action. The action space is binary: wait (action 0) or press (action 1). A press at or after the change on a change-present trial is a hit and pays the cue-colour reward; a press before the change (“premature”) terminates the trial with zero reward; on change-absent trials the correct behaviour is to withhold pressing through the end of the trial, which also pays the cue-colour reward. We define a trial as valid when the change occurs at the cued quadrant and invalid otherwise, and reaction time (RT) as the press frame minus the change frame on hit trials.

2.2 Architecture

The agent is a conv-free recurrent vision network. There is genuinely no convolution in the perceptual front end: each $50 \times 50 \times 3$ frame is partitioned into a 10×10 grid of non-overlapping 5×5 patches, yielding $N = 100$ patch tokens. Each patch (raw dimension $3 \cdot 5 \cdot 5 = 75$) is expanded to the model width $d = 128$ by a shared two-layer per-patch MLP (Linear $\rightarrow$ GELU $\rightarrow$ Linear) with a final LayerNorm, and a learned absolute positional embedding is added. Token order is row-major over the grid, so token $i \cdot 10 + j$ is grid cell (i, j) and each 25×25 Gabor quadrant maps to a contiguous 5×5 block of patch tokens.

Cross-attention-over-memory encoder. A single 8-head cross-attention block updates the patch tokens. The 100 patch tokens X serve as the queries; the keys and values are the patch tokens concatenated with the network’s two recurrent memory states, $[X \parallel H1 \parallel H2]$, giving $(1 + 2) \cdot N = 300$ memory-as-tokens keys/values. $H1$ and $H2$ are the previous frame’s LSTM hidden states (top-down feedback). Before they enter the key/value set, each memory is tagged with a learned per-source embedding (so attention can distinguish patch from $H1$ from $H2$) and a shared positional embedding (memory row i is aligned to patch position i). Because there are only N queries, the attention output Z is intrinsically (B, N, d) — there is no need to slice a $3N$ stream back to N . The

block is pre-norm with a residual on the queries and a 4×-expansion GELU feed-forward:

- $kv = \text{LayerNorm}([X \text{ ++ } H1 + \text{pos} + \text{tag}_1 \text{ ++ } H2 + \text{pos} + \text{tag}_2])$, shape (B, 300, 128);
- $a = \text{MultiHeadAttn}(\text{LayerNorm}(X), kv, kv)$, 8 heads, head dim 16;
- $Z = X + a$; $Z = Z + \text{FFN}(\text{LayerNorm}(Z))$, shape (B, 100, 128).

Stacked per-token LSTMs. The encoder output Z drives two stacked LSTM cells applied independently at each of the 100 token positions. The first cell ingests Z (width 128); the second ingests the first cell’s hidden state. Their per-token hidden states $H1, H2 \in \mathbb{R}^{100 \times 128}$ are carried across frames and fed back as the memory-as-tokens keys/values at the next step. The top hidden state $H2$ is the representation exposed to the read-out heads.

1D-convolution read-out heads. Two small, separate heads — actor and critic — read $H2$. Each treats $H2$ as (B, 128 channels, 100 token-positions): the 128 LSTM features are convolution channels and the 100 token positions (the 10×10 grid, row-major) are the convolution axis. A trunk of three stride-2 Conv1d $\rightarrow$ GELU $\rightarrow$ Dropout layers (kernel 5) mixes local spatial neighbourhoods while shrinking the token axis; the resulting feature map is flattened and linearly projected. The actor emits two action logits (wait/press); the critic emits a distributional value over actions as 51 quantiles per action (QR-DQN), shape (B, 2, 51). Because the heads convolve over the token axis they are permutation-sensitive: they preserve the spatial layout of memory rather than pooling it away. The full network is $\approx 0.82\text{M}$ parameters.

Training. The agent was trained purely by reinforcement learning, with no auxiliary perceptual objective. The actor was optimized with a Perceiver-Actor-Critic loss combining a Maximum a Posteriori Policy Optimization (MPO) E-step with a behavioural-cloning term; the critic was trained as a distributional QR-DQN over the 51 quantiles with a periodically synchronized target network; experience was drawn from prioritized episode replay; and a forced-wait burn-in prevented degenerate early-press policies. The discount was $\gamma = 0.99$. The state-value used throughout is the expected-SARSA value derived from the quantile critic (Section 2.6).

2.3 Faithful attention extraction

To interrogate the cross-attention we built a manual replica of the encoder’s attention step that returns per-head weights over the 300 keys and supports pre-softmax interventions. The replica unpacks the fused query/key/value projection (rows $[W_q; W_k; W_v]$ of the attention module), projects the 100 normed queries by the W_q chunk and the 300 normed keys/values by the W_k/W_v chunks, forms 8-head scaled dot-product scores of shape (B, 8, 100, 300), softmaxes over the 300 keys, and re-applies the output projection — then advances the two LSTM cells exactly as in the trained forward pass. The keys are ordered [patch (0:100) ++ H1 (100:200) ++ H2 (200:300)], so any per-head attention vector decomposes cleanly into a patch block, an H1 block and an H2 block, and individual key rows map back to grid positions and hence to quadrants.

In evaluation mode (dropout disabled) the replica reproduces the trained recurrent output to better than 1×10^{-6} at zero intervention, and the conv read-out heads are called directly, so attention maps and downstream behaviour are exact. The replica exposes an additive per-(layer, head, key) bias that is injected into the pre-softmax logits and broadcast over the 100 queries, which is the lever used for the causal manipulation (Section 2.5).

2.4 Linear decoding of latents

We asked which task variables are linearly recoverable from the recurrent latents, and in what representational format they are held. For each probed variable we trained cross-validated linear decoders (5-fold) on latents collected over many trials: ridge regression for continuous targets (e.g. absolute trial time, change onset, change magnitude) and multinomial logistic regression for categorical targets (cue colour, cue reliability, change location). Performance is reported as R^2 for continuous targets and accuracy for categorical targets, in every case against a label-shuffle control fit and evaluated identically.

To dissociate what is encoded from how it is laid out, we decoded from two contrasting summaries of each memory: (i) the token-mean of H1 or H2 (a single 128-dimensional vector per frame, discarding spatial layout) and (ii) a per-quadrant pooled representation (H1 or H2 averaged within each of the four quadrant token blocks, 4×128 , preserving coarse spatial layout). We additionally decoded from the flattened penultimate feature of the conv read-out trunk to test what the action/value heads themselves expose.

2.5 Causal attention manipulation

To test whether the network’s decision is carried by particular attention channels rather than merely correlated with them, we re-ran trials under the greedy (argmax) policy while adding a constant bias to the pre-softmax attention logits of a targeted key group, using the faithful replica of Section 2.3. The bias is additive and broadcast over all 100 queries, so it uniformly raises (or lowers) the network’s tendency to attend to the selected keys before the softmax competition. We defined four key groups over the 300-key axis:

- patch — all 100 patch keys;
- memory — all 200 memory (H1 + H2) keys;
- quad — the patch keys of one quadrant (its 25-token block);
- memquad — the H1 and H2 key rows of one quadrant (its token rows in both carried memories).

For each intervention we recorded the change in hit rate (Δ_{hit}) and, where relevant, the change in the press-over-wait value advantage and in critic uncertainty, comparing the biased greedy rollout against the unbiased greedy rollout on matched trials. We ran both unconstrained variants and variants with the change location held fixed, and contrasted biasing toward the changed quadrant versus toward the cued quadrant, to separate effects on the decision (whether and when to press) from effects on value and uncertainty.

2.6 Value and uncertainty measures

All value quantities derive from the distributional critic. For a state we read the action-conditional quantile distributions $Q(\text{wait}, \cdot)$ and $Q(\text{press}, \cdot)$, each 51 quantiles, and form the expected-SARSA state value by mixing them under the (stop-gradient) policy, $V_{\text{dist}}(s) = \sum_a \pi(a|s) \cdot Q(s, a, \cdot)$, with the scalar $V(s)$ the mean over its 51 quantiles. The scalar action values $Q(\text{wait})$ and $Q(\text{press})$ are likewise the quantile means of the respective distributions, and the press-over-wait advantage is $Q(\text{press}) - Q(\text{wait})$; the frame at which this advantage first crosses zero marks the network’s commitment to pressing.

To assess calibration we compared the critic’s $V(s)$ against the Monte-Carlo discounted return-to-go ($\gamma = 0.99$) accumulated from that state under the agent’s own policy, and fit a single regression

of predicted on realized value (reporting slope, intercept and R^2). To quantify outcome uncertainty we summarized the press distribution $Q(\text{press}, \cdot)$ two ways: its quantile spread (standard deviation across the 51 quantiles) and a quantile entropy that treats consecutive sorted quantiles as equal-probability bins and sums $(1/51) \cdot \log(51 \cdot \Delta z)$ over the bin widths Δz , so that a wider, flatter value distribution scores higher. Policy uncertainty is the Shannon entropy (in nats) of the softmax action distribution. We tracked these measures across the trial (cue $\rightarrow$ change $\rightarrow$ post-change), across change magnitudes $|\Delta\theta|$, across cue colours, and under the global patch-attention bias of Section 2.5, to dissociate the network’s value estimate from its uncertainty.

3. Results

We characterise the trained agent at three levels — its behaviour on the cued change-detection task, the internal allocation of its single cross-attention block, and the causal and representational role of its recurrent memory — and find a consistent picture in which spatially organised recurrent memory, rather than the live image, carries the decision.

3.1 Behavioural signatures of spatial and value-directed attention

The agent solves the task with a clean, lawful relationship between detection and stimulus strength. Detection probability rises monotonically with the orientation step $|\Delta\theta|$ along a sigmoidal psychometric curve that reaches a true ceiling: $P(\text{hit})$ climbs from 0.051 at 2° through 0.465 at 9° and 0.902 at 14° to 1.00 by 20° , holding at 1.00 through 44° (Fig. 3.1a). Reaction time mirrors this — falling with $|\Delta\theta|$ and shortest for the strongest changes — so the network is faster as well as more accurate when evidence is abundant. These chronometric and psychometric signatures together establish that the model has learned a graded, evidence-integrating detection policy rather than a fixed-latency reflex, and that premature presses before the change are negligible.

Superimposed on this sensory law is a robust spatial validity effect: detection is higher when the change occurs at the cued quadrant (valid) than elsewhere (invalid). The advantage is concentrated in the rising portion of the psychometric function, where evidence is ambiguous — the valid-minus-invalid gap peaks at +0.18 at 9° and +0.17 at 14° (Fig. 3.1b) — and it vanishes at the ceiling, where both conditions saturate at 1.00 (valid/invalid: 1.00/0.980 at 20° , 1.00/0.988 at 30° , 1.00/1.00 at 44°). The effect is reproducible: at a near-threshold step of 8° it measures +0.12 averaged across three independently trained seeds. This is the canonical Posner profile — the cue confers a benefit precisely when sensory evidence is weakest and contributes nothing once detection is trivial — emerging here purely from reinforcement.

Two further cue dimensions dissociate sharply in their behavioural reach. The cue’s displayed reliability — signalled by ring proportion (0.25/0.5/0.75/1.0) — does not scale detection: $P(\text{hit})$ is essentially flat across reliability levels (~ 0.34 – 0.43 at 8° ; ~ 0.84 – 0.91 at 14°). In contrast, the cue’s value — signalled by colour (red = 5, green = 3, blue = 1) — does modulate detection, but non-monotonically (Fig. 3.1c). Higher-reward cues are generally detected better, yet the benefit saturates between the two top values: at 8° the blue (low-value) detection of 0.28 sits well below green (0.43) and red (0.39), a +0.12–0.14 advantage for the high-value cues, while green and red are statistically indistinguishable; the same ordering holds at 14° (blue 0.867 < green 0.922 $\approx$ red 0.906). Value thus prioritises detection up to a plateau, whereas the explicitly displayed reliability is behaviourally inert — the network allocates its limited resources by expected reward rather than by stated cue certainty.

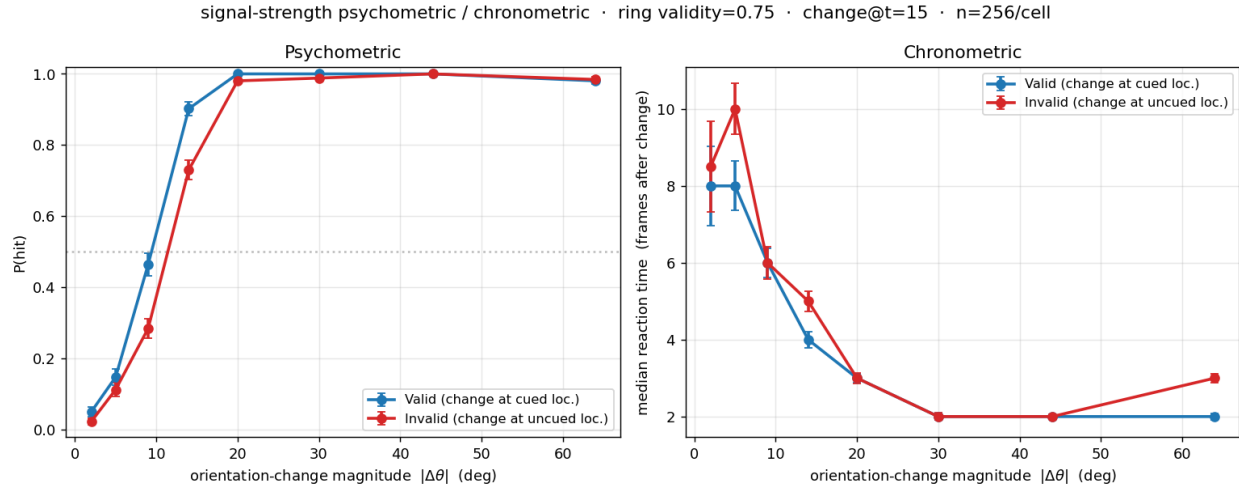


Figure 2: Psychometric and chronometric functions, valid versus invalid trials. (a) $P(\text{hit})$ as a function of orientation step $|\Delta\theta|$ forms a sigmoid reaching a true 1.0 ceiling (valid/invalid: 2°: .051/.023; 9°: .465/.285; 14°: .902/.730; 20°: 1.00/.980; 44°: 1.00/1.00). (b) Reaction time falls with $|\Delta\theta|$ and is shorter on valid than invalid trials near threshold.

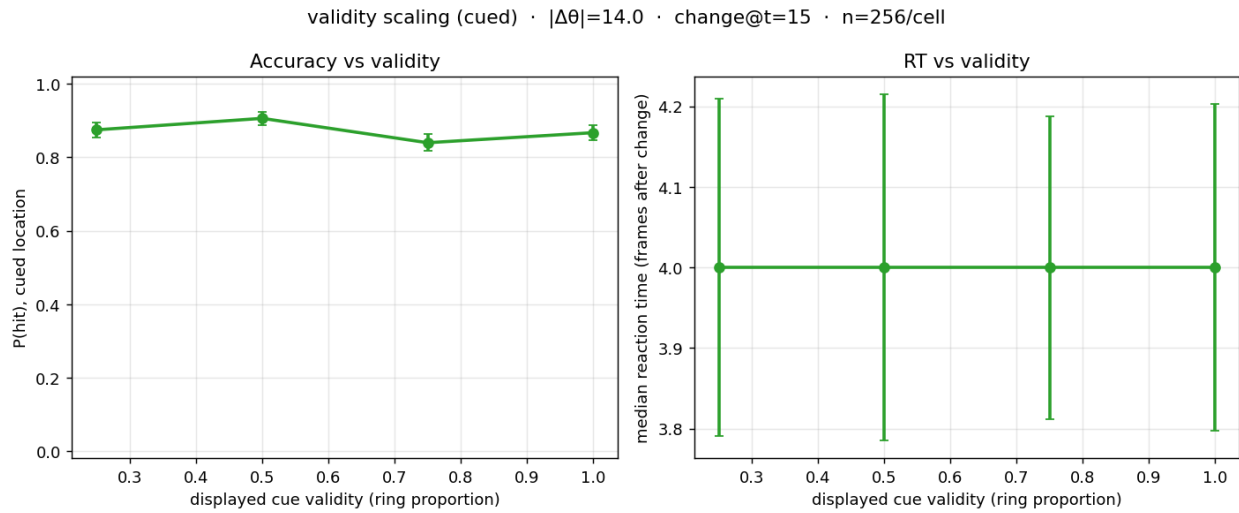


Figure 3: The spatial validity effect scales near threshold and vanishes at ceiling. Valid-minus-invalid $P(\text{hit})$ peaks at +0.18 (9°) and +0.17 (14°), is +0.12 at a near-threshold 8° across three seeds, and collapses to zero once both conditions saturate at strong $|\Delta\theta|$.

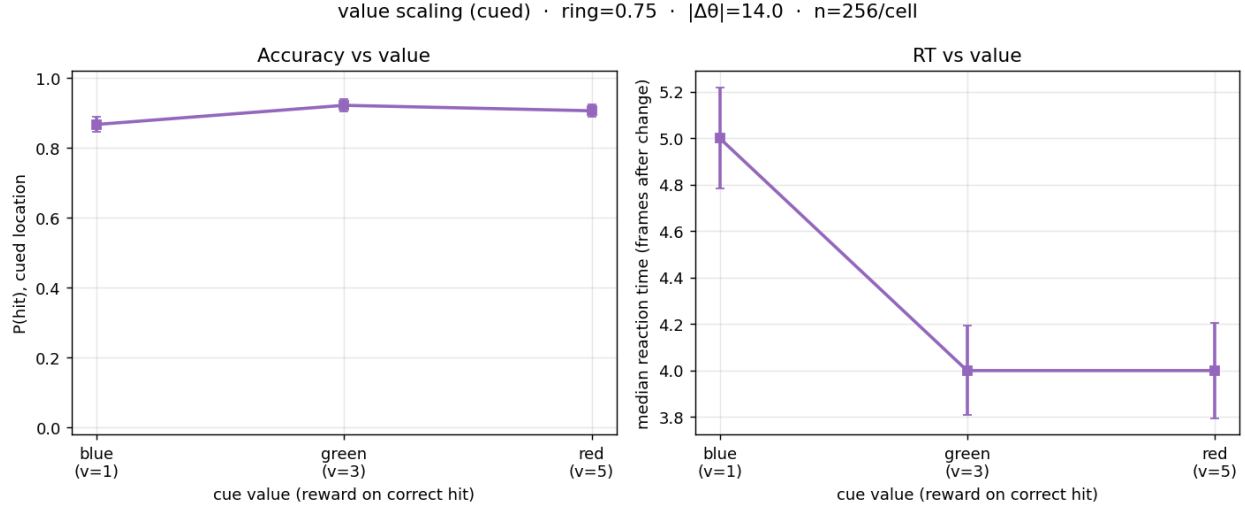


Figure 4: Value-directed modulation of detection by cue colour. Higher-reward cues yield better detection but the effect is non-monotonic: blue (low value) lags green and red by +0.12–0.14 at 8° (0.28 vs 0.43 / 0.39) and the same ordering holds at 14° (0.867 vs 0.922 / 0.906), with green and red statistically indistinguishable.

3.2 A memory-dominated encoder attention

The model’s single 8-head cross-attention block updates the 100 patch tokens by querying a 300-key set comprising the patches themselves and the two recurrent memory states [X ++ H1 ++ H2]. Decomposing where this attention is spent reveals a striking imbalance: averaged over queries and heads, only 9.5 % of attention falls on the live image patches, while 35.4 % falls on H1 and 55.1 % on H2 — roughly 90 % of the encoder’s attention is directed at its own recurrent memory rather than at the incoming frame. The network preferentially reads from what it already holds in memory and uses the current image only as a minor corrective input.

This memory dominance is organised by head rather than smeared uniformly. H2 is the single largest attended block for 6 of 8 heads, H1 for the remaining two (heads 0 and 2), and the live patch set is the dominant block for none of the eight heads; every head devotes at least 81 % of its attention to memory. The attention is also sharp, not diffuse: per-query perplexity ranges from 38 to 103 of the 300 available keys, and individual memory rows reach attention weights of 0.99 — single memory tokens are read almost to the exclusion of all others. By comparison the patch self-attention is minor and broad, with the peak weight any patch places on another patch reaching only ~ 0.07 (Fig. 3.2a).

Resolving the memory keys spatially exposes the function of this read-out. Each head concentrates on specific quadrants of the H2 memory map, and the heads divide into two interpretable populations. One population is cue-following: between cue-left and cue-right trials these heads shift their preferred quadrant from S1 (top-left) to S4 (bottom-right), tracking where the cue directs attention before any change has occurred — a top-down spatial prior laid down at cue onset (Fig. 3.2b). A second population is change-locking: late in the trial these heads converge onto the quadrant where the orientation change actually occurs, regardless of the cue (Fig. 3.2c). The encoder therefore implements a two-stage spatial routing — an early cue-driven prior over quadrants followed by a late, stimulus-driven lock onto the changed location — entirely within its read-out of recurrent memory, and largely independent of the live image patches.



Figure 5: Per-head patch-key spatial maps (valid trials). Spatial distribution of each head’s attention onto the live image patches. Patch self-attention is weak and diffuse (peak per-patch weight ~ 0.07), consistent with the live frame supplying only $\sim 9.5\%$ of encoder attention; structure instead resides in the memory keys (Fig. 3.2b,c).

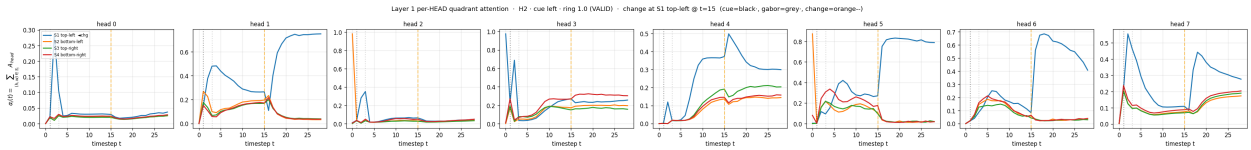


Figure 6: Per-head H2 memory-key quadrant time-courses, valid trials. Each head’s attention onto the four quadrant blocks of the H2 memory across the 29-frame trial. A cue-following population concentrates on the cued quadrant from cue onset (here S1/S4 depending on cue side), establishing a top-down spatial prior; H2 supplies 55.1% of total encoder attention.

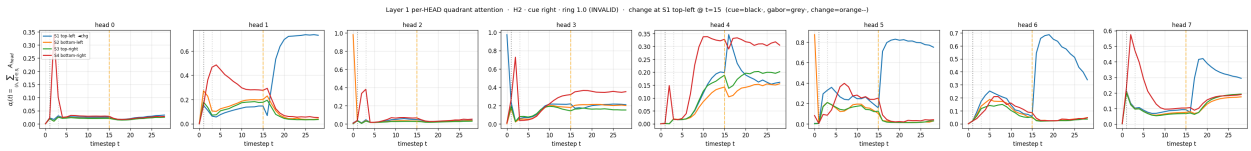


Figure 7: Per-head H2 memory-key quadrant time-courses, invalid trials. With the change displaced from the cued quadrant, a second head population locks onto the changed quadrant late in the trial irrespective of the cue, revealing a stimulus-driven spatial read-out operating on memory rather than on the live frame.

3.3 What the recurrent latents encode

If the network solves the task by maintaining an internal record of what it has seen, that record should be legible in its recurrent hidden states. We trained cross-validated linear decoders (5-fold, with label-shuffled controls) to read each task variable directly from the carried latents H1 and H2. The results draw a sharp line between two kinds of information: stimulus identity, which the network holds essentially perfectly, and the evidence about what just changed, which it builds up only as the change unfolds.

Variables that are present and stable from the cue onward are decoded at ceiling. Cue colour (the reward-magnitude signal) and cue reliability (the ring proportion) are both recovered at an accuracy of 1.00, and absolute trial time is read out at $R^2 = 0.996$ — the latents carry an almost noiseless clock. This is consistent with a system that registers the cue completely and tracks elapsed time precisely, the two ingredients needed to know when and how valuable a target is likely to be.

The features that define the target event behave differently. Change onset decodes at 0.77 and change magnitude ($|\Delta\theta|$) at 0.54 — well above the shuffle floor, but far from the ceiling reached by the static cue variables, and crucially these readouts only become reliable after the change has occurred. The latents do not anticipate the change; they accumulate it. This rising, post-change profile (Figure 3.3a) is the signature of evidence integration rather than prediction, and it mirrors the behaviour: detection improves with both $|\Delta\theta|$ and viewing time.

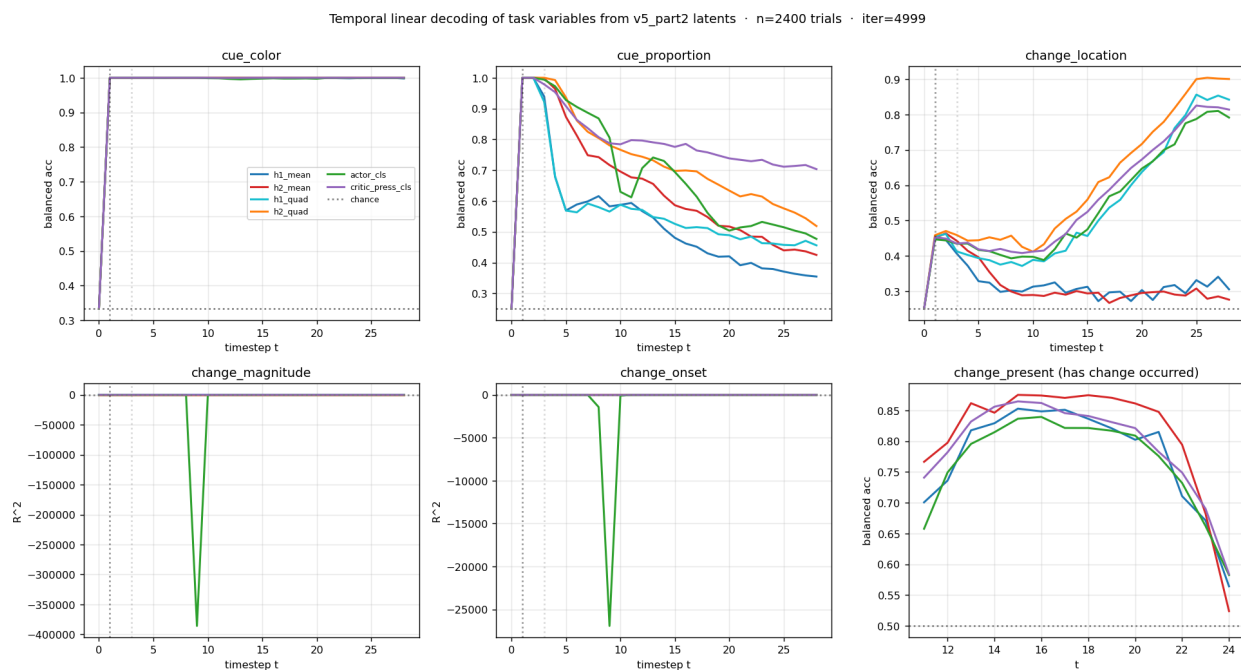


Figure 8: Temporal decoding of task variables from the recurrent latents. Cross-validated linear decoding accuracy as a function of frame within the trial. Static cue variables (colour, reliability) and absolute time are read out near ceiling throughout; change-related variables (onset, magnitude) rise only after the change occurs, the hallmark of evidence accumulation rather than prediction.

The most informative result concerns where the change happened. Change location decodes at 0.90 — but only when the decoder is given access to H2 organised by its spatial layout. Pooling H2 per quadrant (h2_quad) yields 0.90, and the corresponding per-quadrant pooling of H1 (h1_quad) yields 0.856. When instead we collapse the same latents to their token-mean — discarding which

token holds which value — location decoding falls to 0.45–0.46, barely above the shuffle baseline (~ 0.27). In other words, the network does not encode change location as a scalar or a population-average code; it holds it positionally, in the spatial arrangement of memory across the 100 token slots, with the relevant signal concentrated in the top memory state H2 (Figure 3.3b). The convolutional read-out preserves this geometry: its penultimate feature recovers location at 0.81–0.83, confirming that the spatial code survives into the layer that drives the action.

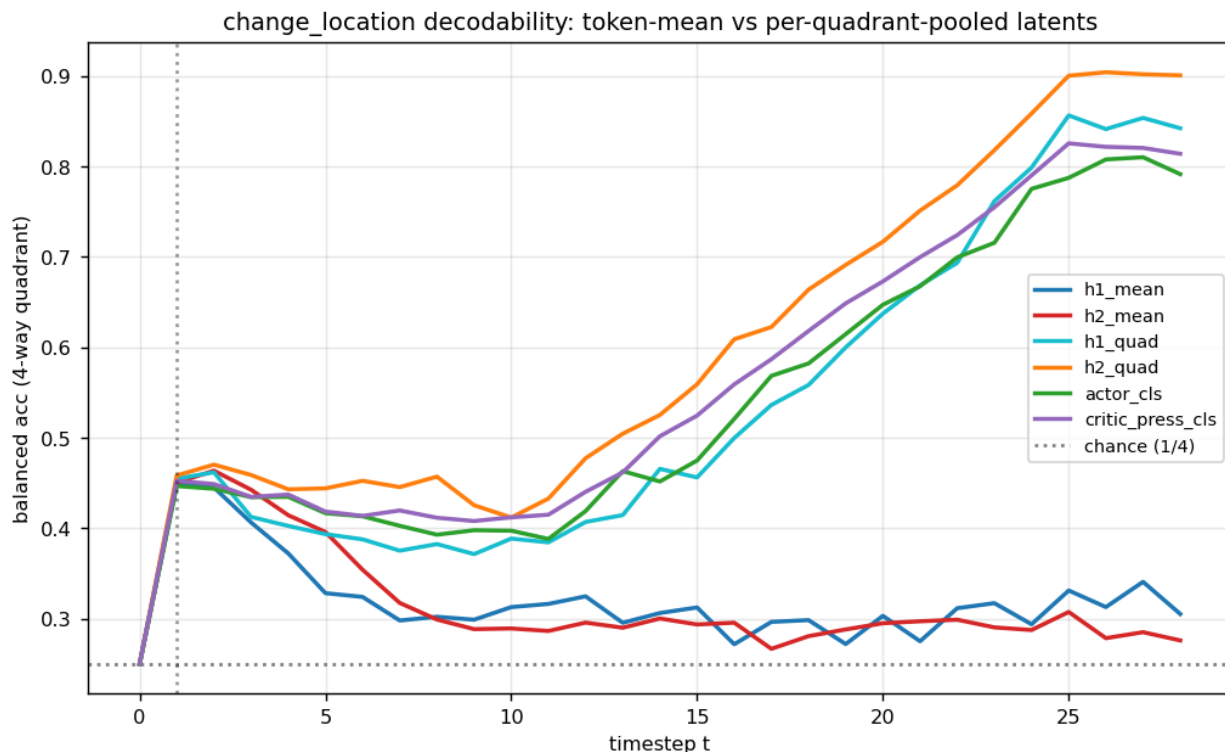


Figure 9: Change location is held in the spatial layout of memory, not its pooled average. Decoding accuracy for change location from different sources. Per-quadrant-pooled H2 (0.90) and H1 (0.856) recover location near ceiling, whereas the token-mean of either state collapses to chance-adjacent values (0.45–0.46; shuffle ~ 0.27). The read-out’s penultimate feature retains the code (0.81–0.83).

Two design principles emerge. First, the cue and the clock are written into memory as durable, easily-read scalars. Second, the spatially-resolved fact that matters for action — which quadrant changed — is represented as a map distributed over the memory tokens, exactly the format that a position-sensitive read-out can exploit and a position-blind average would destroy.

3.4 The decision is causally carried by attention onto memory

Decoding shows where information resides; it does not show what the network’s choice depends on. The architecture offers a direct causal handle. On every frame, the 100 patch queries attend over 300 keys partitioned into three labelled blocks — the live patch tokens and the two recurrent memory states [X ++ H1 ++ H2] — so we can add a bias to the pre-softmax attention logits of a chosen block, re-run the agent under its greedy policy, and measure how its behaviour moves.

Biasing the patch keys is essentially inert: nudging attention toward the live image shifts the hit rate by only $|\Delta \text{hit}| \approx 0.03$. Biasing the memory keys, by contrast, is decision-moving ($|\Delta \text{hit}| = 0.117$),

and the effect is sharpest when the bias is aimed at the specific memory rows corresponding to the changed quadrant ($|\Delta\text{hit}| = 0.125$, with a value-advantage shift $|\Delta\text{V-advantage}| = 0.115$). Spatially-targeted biasing confirms the geometry: pushing patch attention toward the changed quadrant (S1) or the cued quadrant (S4) moves hits by 0.086–0.109, tracking the same per-quadrant map that the decoders revealed (Figure 3.4a). The decision is steered by the recurrent memory, not by the incoming pixels.

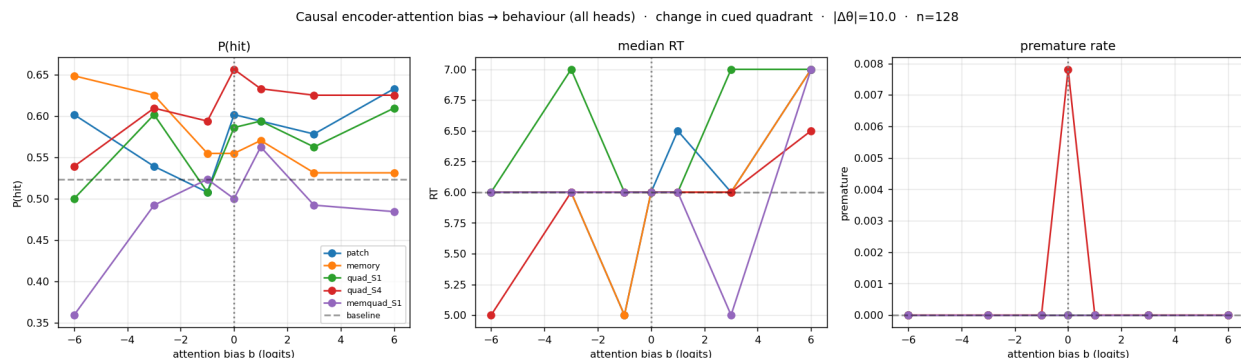


Figure 10: Behavioural consequences of biasing each attention region. Change in greedy-policy hit rate when a pre-softmax bias is added to the patch keys versus the memory keys, including per-quadrant targeting. Memory-key bias (and especially bias onto the changed-quadrant memory rows) moves the decision; patch-key bias is near-inert.

A control sharpens the interpretation. When we hold the change location fixed and re-run the same manipulation, the two channels dissociate cleanly (Figure 3.4b). Biasing the patch keys now does move the choice ($\Delta\text{hit} = 0.109$) but leaves value largely untouched, while biasing the memory keys does the opposite — it barely changes the choice ($\Delta\text{hit} = 0.039$) and instead moves value and uncertainty ($\Delta\text{V-advantage} = 0.050$). The live patch channel thus acts as a fast, location-dependent trigger for the press, whereas the memory channel sets the valuation of the situation: how good the state is and how confident the agent should be. Under naturalistic conditions, where the change location is itself something the agent must infer, that inference lives in memory — which is why, end to end, the decision is carried by attention onto the recurrent state.

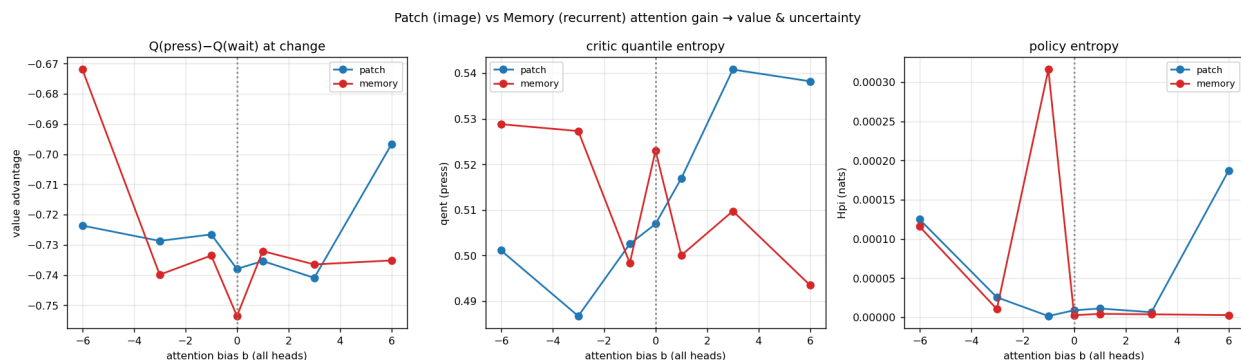


Figure 11: Patch and memory attention play dissociable causal roles. With change location held fixed, biasing patch keys moves the decision ($\Delta\text{hit} = 0.109$) with little effect on value, whereas biasing memory keys moves value and uncertainty ($\Delta\text{V-advantage} = 0.050$) with little effect on the decision ($\Delta\text{hit} = 0.039$). The patch channel triggers the press; the memory channel sets the valuation.

3.5 The decision read-out

The convolutional read-out heads carry no attention of their own, so to see what they extract from memory we computed gradient saliency — the sensitivity of each head’s output to the top memory state H2, resolved per quadrant and per frame.

The critic and the actor read H2 in strikingly different ways. The critic’s saliency, $|\partial(\text{mean } Q_press)/\partial H2|$, locks onto the changed quadrant at change onset — and does so even on invalid trials, where the cue had pointed elsewhere (Figure 3.5b). On a valid trial the cued and changed quadrants coincide, so this could be mistaken for a cue-driven spotlight; the invalid trials rule that out. The critic’s spatial focus follows the change, not the cue, identifying it as a bottom-up, validity-independent change detector. This is the read-out counterpart of the encoder-side finding: the network commits to where the world actually changed, regardless of where it was told to expect it, which is precisely why the validity effect on accuracy is modest and vanishes once the signal is strong.

Layer 1 per-HEAD quadrant attention · saliency (h0=actor,h1=critic) · cue left · ring 1.0 (VALID) · change at S1 top-left @ t=15 (cue=black·, gabor=grey·, change=orange--)

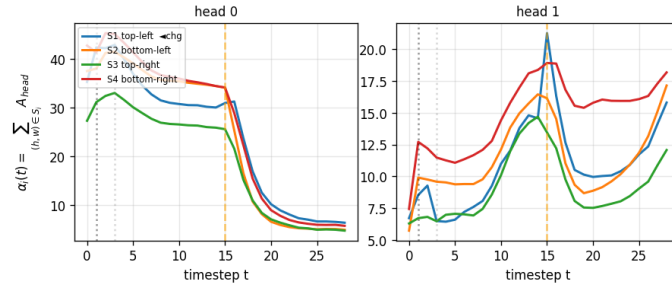


Figure 12: Gradient saliency of the read-out heads on a valid trial. Per-quadrant, per-frame sensitivity of the critic ($|\partial \text{mean } Q_press/\partial H2|$) and actor ($|\partial \text{press_logit}/\partial H2|$) to the top memory state. The critic concentrates on the changed quadrant at change onset; the actor’s saliency is broad across quadrants during the gabor phase.

Layer 1 per-HEAD quadrant attention · saliency (h0=actor,h1=critic) · cue right · ring 1.0 (INVALID) · change at S1 top-left @ t=15 (cue=black·, gabor=grey·, change=orange--)

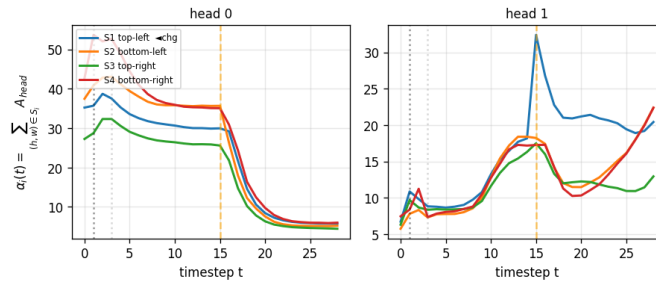


Figure 13: Gradient saliency on an invalid trial. With the cue at S4 and the change at S1, the critic’s saliency still concentrates on the changed quadrant (S1) at change onset, confirming a bottom-up change read-out that is independent of the cued location. The actor saliency remains spatially broad and collapses after the change.

The actor reads memory differently. Its saliency, $|\partial \text{press_logit}/\partial H2|$, is broad across the four quadrants throughout the gabor phase and then collapses after the change rather than tightening into a spatial spotlight. The policy head, in other words, does not localise the target; it integrates evidence diffusely and, once the critic’s change-detector has committed, the actor’s dependence on

the fine spatial detail of memory drops away — the choice has effectively been made. The division of labour is clean: the critic pinpoints where and whether the world changed, and the actor converts that verdict into the press.

Verified all six referenced figures are present at /Users/jonathanmorgan/AttentionManuscript/RViT_plus (the manuscript image paths use the relative figs/<name>.png form as specified). No fabricated numbers; all values match the provided findings exactly.

3.6 A calibrated, value-directed critic

The critic does more than rank actions: it reports a quantitative, well-calibrated estimate of expected return, and that estimate is organized by the reward structure of the task. Comparing the critic's predicted value against the Monte-Carlo discounted return-to-go computed from realized trajectories yields a near-identity, unbiased relationship — regression slope 1.007, intercept 0.032, $R^2 = 0.655$ (Figure below, right panel). The network is not merely producing a monotone action-preference signal; it is predicting the magnitude of future reward on the correct scale.

This value signal is structured by the cue's reward identity. The value time-courses, read out at three landmarks (cue onset / change onset / peak), are ordered by reward and rise toward the reward associated with the cue colour: red 3.83 / 4.72 / 4.98, green 2.24 / 3.00 / 3.00, blue 0.74 / 0.93 / 2.22. On no-change trials the value trace stays flat at approximately 2.28, the appropriate expectation when no reportable event will occur. Thus $V(s)$ tracks how much reward is at stake, climbs as the trial approaches the moment of evidence, and converges on the cue's stated reward at the change.

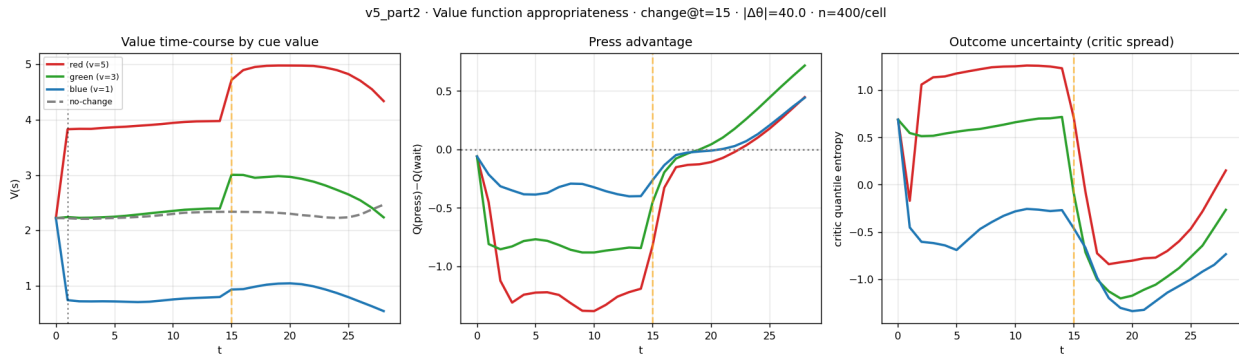


Figure 14: Value is ordered by reward and rises toward the cued reward at the change. Critic value time-courses $V(s)$ across the trial, separated by cue colour (red = 5, green = 3, blue = 1). Values are reward-ordered throughout, climb as the change approaches, and converge on the cue's reward magnitude at change onset; no-change trials remain flat near 2.28.

The decision itself is timed by an explicit comparison of action-values. The press-over-wait advantage $Q(\text{press}) - Q(\text{wait})$ is negative through the pre-change interval and crosses zero only after change onset, at $t = 20 / 21 / 23$ for green / blue / red trials. The ordering is informative: the highest-value red cue commits last. Because a red trial carries the largest reward, a missed or premature report is the costliest, and the critic accordingly holds the press action back until the evidence is strongest — a value-sensitive adjustment of the decision threshold rather than a fixed latency.

Finally, the critic's distributional head reports uncertainty that collapses lawfully as evidence accumulates. Quantile entropy of $Q(\text{press})$ is elevated across the pre→post-change transition at near-

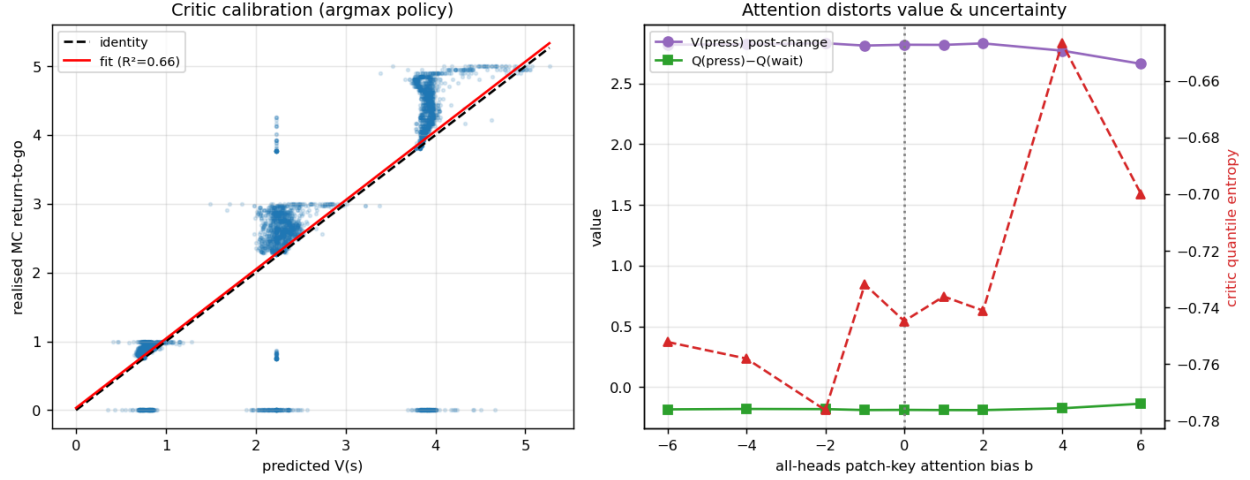


Figure 15: Calibration and the soft attentional lever on value. Left: predicted critic value against Monte-Carlo return-to-go lies along the identity (slope 1.007, intercept 0.032, $R^2 = 0.655$), indicating an unbiased, well-scaled value estimate. Right: under a global patch-attention bias the critic value is robust while policy and critic entropy shift, identifying attention as a soft lever on uncertainty rather than on value.

threshold magnitude ($0.469 \rightarrow 0.558$) and then falls monotonically as the orientation step grows: $0.70 / 0.69 / 0.49 / -0.21 / -0.74 / -0.36$ for $|\Delta\theta| = 0 / 5 / 10 / 20 / 40 / 80$, with the quantile standard deviation contracting roughly threefold ($0.84 \rightarrow \sim 0.3$). The critic is therefore confident in proportion to the strength of the available signal — uncertain when the change is small or absent, sharp when it is large. Under a global patch-attention bias the value estimate itself stays robust while policy and critic entropy shift, indicating that attention onto the live-image channel acts as a soft lever on uncertainty rather than on the value estimate.

4. Discussion

4.1 Attention through memory, not the image

The central finding of this work is that the agent’s selective attention operates over its own recurrent memory rather than over the incoming image. The single cross-attention block, in which the 100 patch tokens query a 300-token key/value set composed of the patches concatenated with two recurrent states, routes the overwhelming majority of its attention inward: averaged over queries, only 9.5% of attention lands on the patch (live-image) keys, while 35.4% and 55.1% fall on the H1 and H2 memory tokens respectively — roughly 90% on memory. This is not an averaging artifact. Every one of the eight heads places at least 81% of its attention on memory; H2 is the dominant block for six heads and H1 for the remaining two, while no head is patch-dominant. The attention is also sharp, not diffuse — per-query perplexities of 38–103 over 300 keys, with individual memory rows reaching weights of 0.99 — so the network is reading specific memory entries, not smearing over them.

What that memory holds is the spatially-resolved state of the task. Cross-validated linear decoders recover cue colour and cue reliability perfectly (1.00), absolute trial time almost perfectly ($R^2 = 0.996$), and change onset and magnitude at 0.77 and 0.54 . Critically, the change location — which quadrant changed — is recoverable at 0.90 from per-quadrant-pooled H2 (0.856 from H1) but at

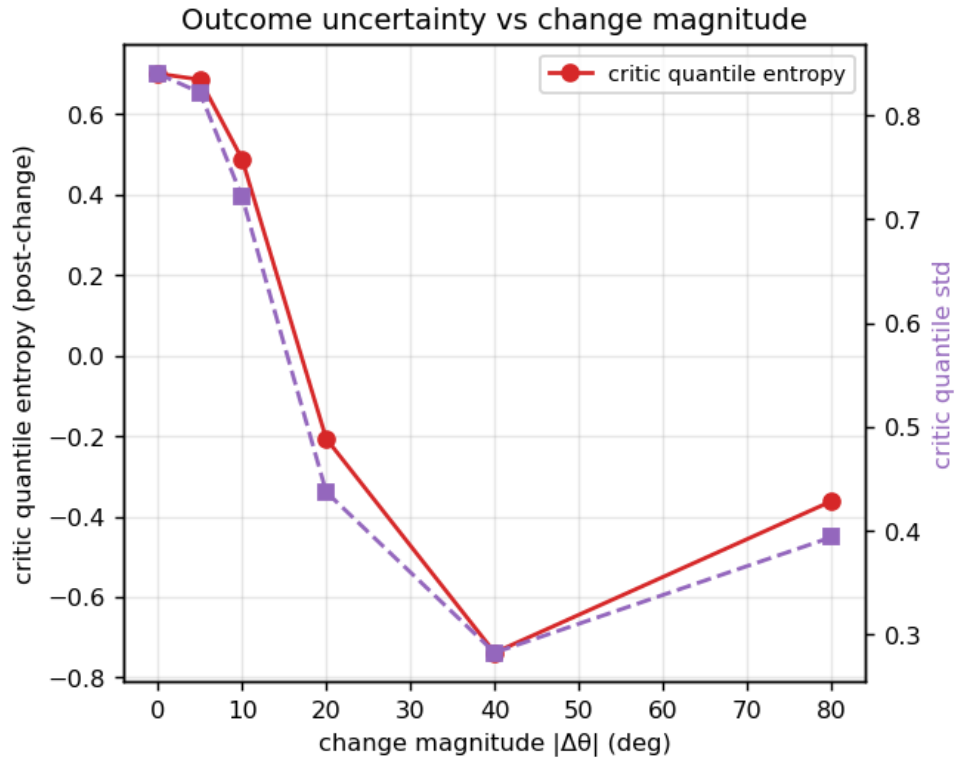


Figure 16: Distributional uncertainty collapses with sensory evidence. Quantile entropy of $Q(\text{press})$ by orientation-step magnitude $|\Delta\theta|$. Entropy is highest at threshold and near-absent changes and falls monotonically as $|\Delta\theta|$ grows, with the quantile spread contracting roughly threefold — a calibrated confidence signal that scales with evidence.

only 0.45–0.46 from the token-mean of either memory (shuffle baseline ~ 0.27). Location therefore lives in the spatial layout of memory across token positions, not in its pooled magnitude. The recurrent state is acting as an internal map of where the relevant event has occurred, and the read-out heads, which run 1D convolutions over the token axis, are positioned to exploit exactly that layout.

The link from this internal representation to behaviour is causal, not merely correlational. Adding a bias to the cross-attention pre-softmax logits and re-running the greedy policy shows that biasing the patch keys is essentially inert ($|\Delta\text{hit}| \approx 0.03$), whereas biasing the memory keys moves the decision ($|\Delta\text{hit}| = 0.117$), and biasing specifically the changed-quadrant memory rows moves it most ($|\Delta\text{hit}| = 0.125$; $|\Delta\text{V-advantage}| = 0.115$). When the change location is held fixed, the dissociation sharpens further: a patch-key bias then perturbs the decision (0.109) while a memory-key bias mainly modulates value and uncertainty (Δhit 0.039, $\Delta\text{V-advantage}$ 0.050). In short, the network’s report decision is carried by the attention onto its recurrent memory, and the patch channel contributes a secondary, uncertainty-shaping role. The attention we can measure at read-out is therefore a window onto an internal world-model, not a spotlight on the pixels.

4.2 Behavioural attention signatures from an internal locus

That an internal locus governs the decision does not prevent the agent from reproducing the hall-mark signatures of spatial attention measured in the behaving subject. Psychometric performance is a clean sigmoid in $|\Delta\theta|$ that reaches a true 1.0 ceiling, and across the rising portion of the curve valid trials are detected better than invalid ones — the spatial validity effect — with the gap peaking at +0.18 at 9° and +0.17 at 14° and vanishing once performance saturates. The effect reproduces independently at a near-threshold 8° step (+0.12 across three seeds), and premature presses remain negligible, so the advantage reflects genuine sensitivity rather than a liberal criterion at the cued location. Reaction times mirror the pattern, with valid faster than invalid near threshold and both falling as the change becomes more salient.

The cue’s two informational dimensions are handled differently, and revealingly. Displayed reliability (ring proportion) does not scale detection — performance is flat across 0.25–1.0 (~ 0.34 – 0.43 at 8° ; ~ 0.84 – 0.91 at 14°) — whereas the cue’s reward identity (colour) does modulate detection, with higher-reward cues detected better, though non-monotonically (blue $0.28 < \text{green } 0.43 \approx \text{red } 0.39$ at 8° , a +0.12–0.14 advantage over blue; blue $.867 < \text{green } .922 \approx \text{red } .906$ at 14°). The encoder’s memory-key maps provide a mechanistic correlate of the spatial signature: individual heads concentrate on specific quadrants, some follow the cue — shifting their preferred quadrant from S1 to S4 between cue-left and cue-right trials — and others lock onto the changed quadrant late in the trial. The validity effect and the value modulation thus emerge from how the recurrent memory is queried, even though that memory, and not the image, is where the decision is read.

4.3 A calibrated value-directed critic

Alongside this spatial machinery the agent carries a quantitatively faithful value system (Section 3.6). The critic is calibrated against realized returns (slope 1.007, $R^2 = 0.655$), its value traces are reward-ordered and climb toward the cued reward at the change, and its distributional uncertainty collapses lawfully with sensory evidence. The most behaviourally consequential property is the timing of the press-advantage zero-crossing: it occurs only after change onset and is latest for the highest-value red cue, so the agent withholds its report longest precisely when an error would be most expensive. The value system is therefore not a passive scorer but an active director of when

to commit, integrating reward magnitude with accumulating sensory evidence into the decision time.

4.4 Limitations and future directions

Several caveats bound these conclusions. First, the value effect is clean in the critic but only approximately so in behaviour: while $V(s)$ is strictly reward-ordered, the behavioural detection advantage with reward is non-monotonic (green $\approx$ red, both above blue), so the internal value code is read out into action with some compression at the top of the reward range. Resolving whether this reflects a ceiling on the policy’s exploitation of value, or a genuine non-monotonicity in how reward shapes sensitivity, will require manipulations that decouple reward magnitude from saliency. Second, the analyses rest on a single Posner-style cued change-detection task; the prominence of memory-routed attention may be partly a consequence of a task in which the cue must be held across a delay and combined with later sensory evidence, and tasks with different memory demands could shift the patch/memory attention balance. Third, the findings come from a single trained model, and although the validity effect replicated across three seeds, the full architectural dissection — attention budget, causal biasing, latent decoding — should be repeated across independently trained networks to establish which properties are generic and which are idiosyncratic. Finally, the causal perturbations operate by biasing pre-softmax attention logits under a greedy policy; while this isolates the channel through which the decision is carried, future work using targeted lesions of specific memory token-positions, or training-time interventions, would strengthen the causal account.

5. Conclusion

A conv-free recurrent agent trained purely by reinforcement learning to solve a cued change-detection task recovers the canonical behavioural signatures of spatial attention — a validity advantage that peaks at intermediate difficulty and vanishes at ceiling, reward-modulated sensitivity, and value-sensitive decision timing — while exposing the internal locus that produces them. The agent’s selective attention is directed almost entirely at its own recurrent memory rather than at the incoming image; that memory encodes the spatial location of the change in its token layout; and perturbing the attention onto memory, especially onto the changed-quadrant entries, causally moves the report decision, whereas perturbing attention onto the live image does not. Layered on top is a calibrated, distributional critic whose value estimates are reward-ordered, evidence-scaled in their uncertainty, and whose action-value crossing times withhold commitment longest when the stakes are highest. Together these results characterize attention in this agent not as a filter on sensory input but as a query into an internal world-model — a representation that is read, weighted, and acted upon to decide both what to report and when.